Quiz 8

1. Consider a ball falling down subject only to the force of gravity, as shown in Fig. 1.

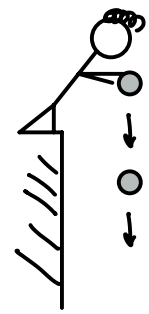

Figure 1: A ball falling vertically downwards

   (a) What is the Lagrangian for the following system? Write this in terms of the ball's position y and velocity $\dot{y}$.

$$\mathcal{L} = T - U$$

$$\mathcal{L} = \frac{1}{2}m\dot{y}^2 - mgy$$

   (b) What are the Euler-Lagrange equations that describe the motion of the ball?

$$\frac{\partial \mathcal{L}}{\partial y} = \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{y}}$$ ← full credit for this

$$-mg = \frac{d}{dt}\left[m\dot{y}\right]$$

$$-mg = m\ddot{y}$$

$$-g = \ddot{y}$$

2. Consider the length of a general curve

$$L = \int_A^B ds$$

Let this curve constrained to a flat 2D plane. Only consider curves that can be written as $y(x)$.

   (a) Rewrite ds in terms of dx and dy

$$ds = \sqrt{dx^2 + dy^2}$$

   (b) Fill in the blank

$$L = \int_{x_A}^{x_B} \underline{\sqrt{1 + y'(x)^2}} \, dx$$